

The 2024 revision of the Declaration of Helsinki: a modern ethical framework for medical research

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Abstract

The Declaration of Helsinki, established in 1964, remains a foundational document in medical research ethics. This review examines the 2024 revision, endorsed by the 75th World Medical Association (WMA) Assembly, highlighting its impact on modern clinical research. Major updates include the shift from “subjects” to “participants,” promoting autonomy and active involvement, and the introduction of dual ethical review requirements for cross-border studies to strengthen accountability. New guidelines for data privacy address AI-related ethical concerns, while enhanced community engagement fosters transparency and shared decision-making. Additionally, standards for environmental sustainability encourage research practices that minimize ecological impacts. In response to global health crises such as COVID-19, the revised Declaration sets forth ethical protections to balance participant safety with research urgency during emergencies. Despite these advances, areas for improvement remain, especially in AI ethics, emergency research protocols, and the extension of the Declaration’s scope to include forensic and specimen research. The 2024 revision thus strengthens the Declaration’s role as an adaptive, relevant framework for safeguarding participant rights and research integrity in a changing landscape.

Keywords: Declaration of Helsinki; medical ethics; participant protection

Introduction: background and need for revision

The Declaration of Helsinki, first adopted in 1964, was established in response to documented ethical violations in medical research, such as those revealed during the Nuremberg Trials following World War II. These revelations highlighted the urgent need for international ethical standards to protect human participants and uphold their rights. Since its inception, the Declaration has become a cornerstone of global research ethics, shaping regulations and guiding research practices across varied cultural and regulatory contexts. Many countries have also used the Declaration as a foundational document for defining or regulating the ethical conduct of medical research.

However, as a “living document,” the Declaration requires periodic updates to address emerging ethical complexities stemming from technological advances and evolving societal expectations. The 2024 revision reflects the need to adapt to significant changes in the clinical research landscape, such as the globalization of clinical trials and the rapid growth of genomics, machine learning, digital health, and artificial intelligence (AI) [1, 2]. These advances present new ethical challenges, including those related to data privacy, potential biases in AI-driven research, and the ethical protection of vulnerable populations. Additionally, the expanded role of digital technologies and global health crises, notably the COVID-19 pandemic, has reinforced the need for refined ethical frameworks to sustain public trust and integrity in medical research.

In response, the World Medical Association (WMA) conducted an extensive 30-month consultation process, gathering input from numerous countries and a wide range of stakeholders, including patients, bioethicists, regulatory officials, and leaders from medical societies. This inclusive, consultative approach reflects the WMA’s commitment to transparency, inclusivity, and responsiveness in addressing both historical and emerging ethical challenges [3].

Historical overview of the Declaration of Helsinki revisions

To prevent a recurrence of the unethical medical research practices exposed during World War II, the WMA developed the Declaration of Helsinki as a statement of ethical principles for human-centered medical research. Since its adoption in 1964, the Declaration has undergone several significant revisions, each reflecting advancements in ethical standards and addressing newly identified gaps. The timeline of the revisions to the Declaration of Helsinki is shown in Fig. 1.

Early revisions (1964–89)

The original text emphasized informed consent and a careful risk-benefit assessment, both essential for safeguarding participant rights. The 1975 Tokyo revision introduced a requirement for independent research ethics committee review, a response to documented abuses like the Tuskegee Syphilis Study in the

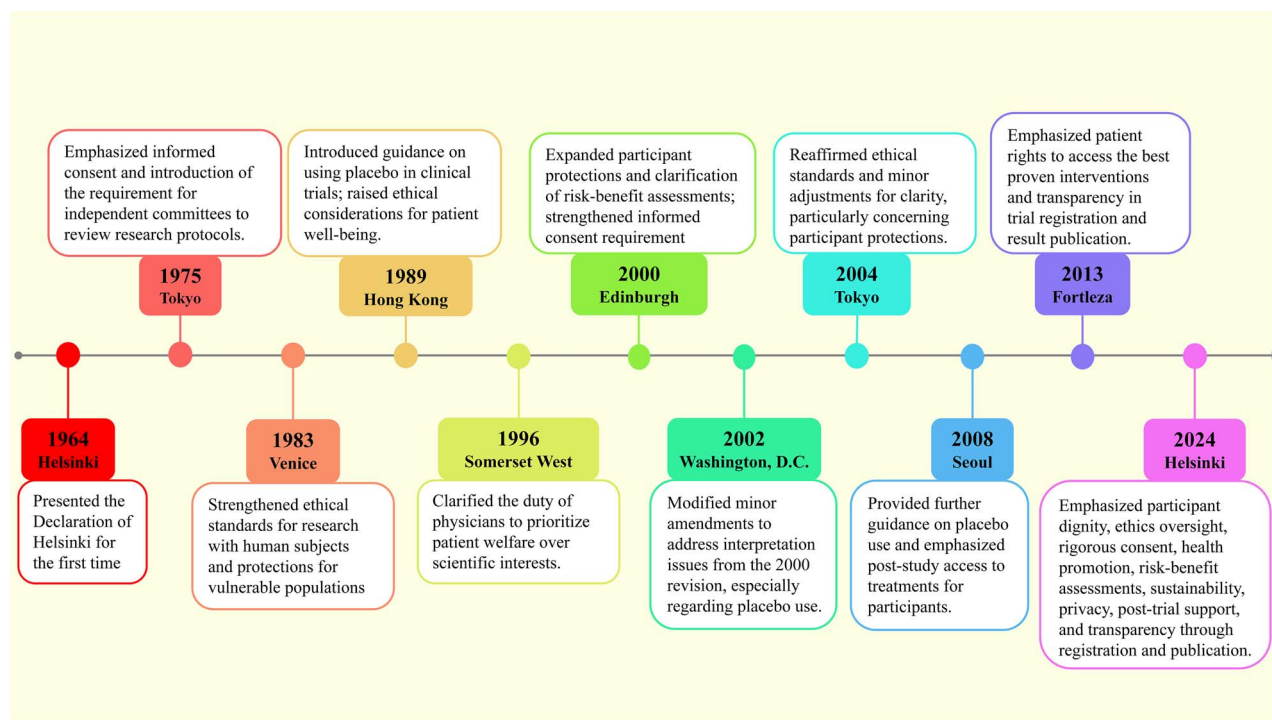


Figure 1 The timeline of the revisions to the Declaration of Helsinki

USA, highlighting the importance of independent oversight to prevent conflicts of interest and exploitation. In 1983, the Venice amendment focused on transparency and disclosure, requiring researchers to clearly communicate research procedures and potential risks to participants. The 1989 Hong Kong revision further reinforced informed consent by mandating written consent for clinical trials and enhancing privacy protections for participants. Collectively, these early revisions aimed to establish consistent standards for informed consent and ethical oversight, setting a global benchmark for participant protection [4].

Globalization and ethical diversity (1996–2008)

With the 1996 revision in Somerset West, South Africa, the Declaration began addressing ethical challenges specific to research in diverse global contexts. This revision underscored the need for consistent ethical principles worldwide, reinforcing protections for vulnerable populations in low-resource settings. The 2000 Edinburgh amendment refined regulations on placebo use, prohibiting placebos without therapeutic benefit when effective treatments were available, thereby upholding participants' rights to the best available care. Additional clarifications on placebo use and ethical standards for multicenter trials were made in Washington D.C. in 2002. The 2004 Tokyo amendment further revisited standards for international trials, emphasizing ethical consistency across borders. The 2008 Seoul revision introduced specific guidelines to ensure that participants in clinical trials receive post-trial benefits, aiming to prevent exploitation in global health research and to ensure that vulnerable groups gain lasting health benefits from participation [5, 6].

Modernization and public health focus (2013–24)

The 2013 revision in Fortaleza, Brazil, marked a pivotal shift toward addressing structural inequalities and providing enhanced protections for vulnerable populations, such as individuals from low socioeconomic backgrounds or those with limited healthcare

access. Building on these foundations, the 2024 revision of the Declaration further expands its ethical scope to include guidance on digital health, data privacy, and environmental sustainability, reflecting the WMA's recognition of the interconnected nature of modern research and global health concerns. This forward-looking revision prepares the Declaration to address current and emerging ethical challenges, particularly considering the growing impact of AI and the need for ecological responsibility within scientific practices [7].

Key 2024 revisions and their ethical implications

Comparison of 2024 and 2013 Helsinki Declaration Texts is shown in Table 1.

Terminological shift from “subjects” to “participants”

The 2024 revision replaces the term “subjects” with “participants,” marking a conceptual shift that recognizes research participants as active contributors rather than passive recipients. This change underscores respect for participants' autonomy, encouraging their involvement in research design and decision-making processes. The new terminology also aligns with recent ethical perspectives on patient-centered research, as seen in studies where patients with chronic diseases actively influence the research process based on their lived experiences rather than being involuntarily subjected to a series of experiments [8].

Enhanced role of ethics committees and researcher qualifications

The revised Declaration mandates dual ethical review in both sponsoring and host countries for cross-border research, addressing the complexities of globalized studies and ensuring ethical accountability at multiple levels. The enhanced qualifications

Table 1. Comparison of 2024 and 2013 Helsinki Declaration Texts.

	2024 Version text	2013 Version text	Difference analysis and significance of changes
1	The World Medical Association (WMA) has developed the Declaration of Helsinki as a statement of ethical principles for medical research involving human participants , including research using identifiable human material or data	The World Medical Association (WMA) has developed the Declaration of Helsinki as a statement of ethical principles for medical research involving human subjects , including research on identifiable human material and data	In the 2024 version, "human participants" replaces "human subjects," which reflects a shift toward more inclusive and respectful language. More emphasis was placed on participant autonomy and motivation rather than passive acceptance. The 2024 version broadens the scope of responsibility beyond physicians to include all parties involved in medical research, reflecting a more inclusive approach to ethical obligations
2	While the Declaration is adopted by physicians, the WMA holds that these principles should be upheld by all individuals, teams, and organizations involved in medical research, as these principles are fundamental to respect for and protection of all research participants, including both patients and healthy volunteers	Consistent with the mandate of the WMA, the Declaration is addressed primarily to physicians. The WMA encourages others who are involved in medical research involving human subjects to adopt these principles	The 2024 text strengthens the ethical obligation by explicitly emphasizing "health and well-being" and the physician's "commitment to the primacy" of these values, reinforcing a more comprehensive patient-centered approach
3	The WMA Declaration of Geneva binds the physician with the words, "The health and well-being of my patient will be my first consideration," and the WMA International Code of Medical Ethics declares, "The physician must commit to the primacy of patient health and well-being and must offer care in the patient's best interest."	The Declaration of Geneva of the WMA binds the physician with the words, "The health of my patient will be my first consideration," and the International Code of Medical Ethics declares, "A physician shall act in the patient's best interest when providing medical care."	No change
4	It is the duty of the physician to promote and safeguard the health, well-being, and rights of patients, including those who are involved in medical research. The physician's knowledge and conscience are dedicated to the fulfillment of this duty	It is the duty of the physician to promote and safeguard the health, well-being, and rights of patients, including those who are involved in medical research. The physician's knowledge and conscience are dedicated to the fulfillment of this duty	No change
5	Medical progress is based on research that ultimately must include participants. Even well-proven interventions should be evaluated continually through research for their safety, effectiveness, efficiency, accessibility, and quality	Medical progress is based on research that ultimately must include studies involving human subjects	The 2024 text replaces "studies involving human subjects" with "participants," streamlining the language and reflecting a broader, more inclusive approach. It also brings forward parts of Article 6, emphasizing the priority of participants' interests
6	Medical research involving human participants is subject to ethical standards that promote and ensure respect for all participants and protect their health and rights. Since medical research takes place in the context of various structural inequities, researchers should carefully consider how the benefits, risks, and burdens are distributed. Meaningful engagement with potential and enrolled participants and their communities should occur before, during, and following medical research. Researchers should enable potential and enrolled participants and their communities to share their priorities and values; to participate in research design, implementation, and other relevant activities; and to engage in understanding and disseminating results	The primary purpose of medical research involving human subjects is to generate knowledge to understand the causes, development, and effects of diseases and to improve preventive, diagnostic, and therapeutic interventions (methods, procedures, and treatments). Even the best proven interventions must be evaluated continually through research for their safety, effectiveness, efficiency, accessibility, and quality	Former Article 6 was split into two parts, with the second part placed in the current Article 5 and the first part in the current Article 7; the content of the former Article 7 (ensuring the interests of participants) has been brought into the present Article; and the addition of a specific interest (fairness, the expression of genuine involvement of the research participants, including in the design, implementation, and understanding and dissemination of the results, etc.), which introduces entirely new content
7	The primary purpose of medical research involving human participants is to generate knowledge to understand the causes, development, and effects of diseases; improve preventive, diagnostic, and therapeutic interventions; and ultimately to advance individual and public health. These purposes can never take precedence over the rights and interests of individual research participants	Medical research is subject to ethical standards that promote and ensure respect for all human subjects and protect their health and rights	The original version of Articles 6 and 8 has been consolidated and the research objectives have been expanded to include the ultimate improvement of human health, both individual and public

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Table 1. Continued

	2024 Version text	2013 Version text	Difference analysis and significance of changes
8	While new knowledge and interventions may be urgently needed during public health emergencies, it remains essential to uphold the ethical principles in this Declaration during such emergencies	While the primary purpose of medical research is to generate new knowledge, this goal can never take precedence over the rights and interests of individual research subjects	Clause 8 in the 2024 version underscores the importance of adhering to ethical principles even in public health emergencies, marking an additional ethical safeguard relevant to crisis situations
9	It is the duty of physicians who are involved in medical research to protect the life, health, dignity, integrity, autonomy , privacy, and confidentiality of personal information of research participants. The responsibility for the protection of research participants must always rest with physicians or other researchers and never with the research participants, even though they have given consent	It is the duty of physicians who are involved in medical research to protect the life, health, dignity, integrity, right to self-determination , privacy, and confidentiality of personal information of research subjects. The responsibility for the protection of research subjects must always rest with the physician or other health care professionals and never with the research subjects, even though they have given consent	The 2024 version shifts to "participants" from "subjects" and replaces "right to self-determination" with "autonomy," reflecting more current ethical terminology, thereby providing greater freedom for participants
10	Physicians and other researchers must consider the ethical, legal, and regulatory norms and standards for research involving human participants in the country or countries in which the research originated and where it is to be performed, as well as applicable international norms and standards. No national or international, legal, or regulatory requirement should reduce or eliminate any of the protections for research participants set forth in this Declaration	Physicians must consider the ethical, legal, and regulatory norms and standards for research involving human subjects in their own countries as well as applicable international norms and standards. No national or international ethical, legal, or regulatory requirement should reduce or eliminate any of the protections for research subjects set forth in this Declaration	The 2024 version includes "other researchers" alongside "physicians" to broaden responsibility. It also specifies both the origin and location of research performance, ensuring comprehensive ethical compliance. The scope of application of the corresponding regulations, norms, and standards has been extended to include both "the country in which the research is initiated and the country in which it is carried out"
11	Medical research should be designed and conducted in a manner that avoids or minimizes harm to the environment and strives for environmental sustainability	Medical research should be conducted in a manner that minimizes possible harm to the environment	The 2024 version introduces the concept of "environmental sustainability," highlighting a broader commitment to sustainable practices in medical research beyond simply minimizing harm. It also requires avoiding environmental damage in experimental design
12	Medical research involving human participants must be conducted only by individuals with the appropriate ethics and scientific education, training, and qualifications. Such research requires the supervision of a competent and appropriately qualified physician or other researcher . Scientific integrity is essential in the conduct of medical research involving human participants. Involved individuals, teams, and organizations must never engage in research misconduct	Medical research involving human subjects must be conducted only by individuals with the appropriate ethics and scientific education, training, and qualifications. Research on patients or healthy volunteers requires the supervision of a competent and appropriately qualified physician or other health care professional	The 2024 version replaces "health care professional" with "researcher" to include broader research roles and emphasizes "scientific integrity" and a stance against "research misconduct," underscoring ethical standards in medical research
13	Groups that are underrepresented in medical research should be provided appropriate access to participation in research	Groups that are underrepresented in medical research should be provided appropriate access to participation in research	No change
14	Physicians who combine medical research with medical care should involve their patients in research only to the extent that this is justified by its potential preventive, diagnostic, or therapeutic value and if the physician has good reason to believe that participation in the research will not adversely affect the health of the patients who serve as research participants	Physicians who combine medical research with medical care should involve their patients in research only to the extent that this is justified by its potential preventive, diagnostic, or therapeutic value and if the physician has good reason to believe that participation in the research will not adversely affect the health of the patients who serve as research subjects	No change

(Continued)

Table 1. Continued

	2024 Version text	2013 Version text	Difference analysis and significance of changes
15	Appropriate compensation and treatment for participants who are harmed as a result of participating in research must be ensured	Appropriate compensation and treatment for subjects who are harmed as a result of participating in research must be ensured	No change
16	In medical practice and in medical research, most interventions involve risks and burdens. Medical research involving human participants may only be conducted if the importance of the objective outweighs the risks and burdens to the research participants	In medical practice and in medical research, most interventions involve risks and burdens. Medical research involving human subjects may only be conducted if the importance of the objective outweighs the risks and burdens to the research subjects	No change
17	All medical research involving human participants must be preceded by careful assessment of predictable risks and burdens to the individuals and groups involved in the research in comparison with foreseeable benefits to them and to other individuals or groups affected by the condition under investigation. Measures to minimize the risks and burdens must be implemented. The risks and burdens must be continuously monitored, assessed, and documented by the researcher	All medical research involving human subjects must be preceded by careful assessment of predictable risks and burdens to the individuals and groups involved in the research in comparison with foreseeable benefits to them and to other individuals or groups affected by the condition under investigation. Measures to minimize the risks must be implemented. The risks must be continuously monitored, assessed, and documented by the researcher	The 2024 version specifies “minimize the risks and burdens,” emphasizing a more detailed approach to managing participant well-being
18	Physicians and other researchers may not engage in research involving human participants unless they are confident that the risks and burdens have been adequately assessed and can be satisfactorily managed. When the risks and burdens are found to outweigh the potential benefits or when there is conclusive proof of definitive outcomes, physicians and other researchers must assess whether to continue, modify, or immediately stop the research	Physicians may not be involved in a research study involving human subjects unless they are confident that the risks have been adequately assessed and can be satisfactorily managed. When the risks are found to outweigh the potential benefits or when there is conclusive proof of definitive outcomes, physicians must assess whether to continue, modify, or immediately stop the study	The 2024 version includes “other researchers” and replaces “human subjects” with “human participants.” It also adds “risks and burdens” to emphasize the comprehensive assessment required
19	Some individuals, groups, and communities are in a situation of more vulnerability as research participants due to factors that may be fixed or contextual and dynamic , and thus are at greater risk of being wronged or incurring harm. When such individuals, groups, and communities have distinctive health needs, their exclusion from medical research can potentially perpetuate or exacerbate their disparities. Therefore, the harms of exclusion must be considered and weighed against the harms of inclusion. In order to be fairly and responsibly included in research, they should receive specifically considered support and protections	Some groups and individuals are particularly vulnerable and may have an increased likelihood of being wronged or of incurring additional harm. All vulnerable groups and individuals should receive specifically considered protection	The 2024 version broadens vulnerability considerations to “individuals, groups, and communities” and introduces the importance of weighing harms of exclusion and inclusion, requiring careful consideration to be given to avoiding experimental and extra-experimental risks to vulnerable groups, enhancing ethical responsibility
20	Medical research with individuals, groups, or communities in situations of particular vulnerability is only justified if it is responsive to their health needs and priorities and the individual, group, or community stands to benefit from the resulting knowledge, practices, or interventions. Researchers should only include those in situations of particular vulnerability when the research cannot be carried out in a less vulnerable group or community or when excluding them would perpetuate or exacerbate their disparities	Medical research with a vulnerable group is only justified if the research is responsive to the health needs or priorities of this group and the research cannot be carried out in a non-vulnerable group. In addition, this group should stand to benefit from the knowledge, practices, or interventions that result from the research	The 2024 version expands the scope to include “individuals, groups, or communities” and emphasizes not exacerbating disparities by exclusion. It provides a more thorough specification of conditions for inclusion more thoroughly, underscoring responsible and inclusive research practices

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Table 1. Continued

	2024 Version text	2013 Version text	Difference analysis and significance of changes
21	Medical research involving human participants must have a scientifically sound and rigorous design and execution that are likely to produce reliable, valid, and valuable knowledge and avoid research waste . The research must conform to generally accepted scientific principles, be based on a thorough knowledge of the scientific literature, other relevant sources of information, and adequate laboratory and, as appropriate, animal experimentation. The welfare of animals used for research must be respected	Medical research involving human subjects must conform to generally accepted scientific principles, be based on a thorough knowledge of the scientific literature, other relevant sources of information, and adequate laboratory and, as appropriate, animal experimentation. The welfare of animals used for research must be respected	The 2024 version emphasizes “rigorous design and execution” and adds the requirement to “avoid research waste,” highlighting the importance of responsible resource use and high-quality research output
22	The design and performance of all medical research involving human participants must be clearly described and justified in a research protocol. The protocol should contain a statement of the ethical considerations involved and should indicate how the principles in this Declaration have been addressed. The protocol should include information regarding funding, sponsors, institutional affiliations, anticipated benefits and potential risks and burdens, methods, anticipated benefits and potential risks and burdens, qualifications of the researcher, sources of funding , any potential conflicts of interest, provisions to protect privacy and confidentiality, incentives for participants, provisions for treating and/or compensating participants who are harmed as a consequence of participation, and any other relevant aspects of the research . In clinical trials, the protocol must also describe any post-trial provisions	The design and performance of each research study involving human subjects must be clearly described and justified in a research protocol. The protocol should contain a statement of the ethical considerations involved and should indicate how the principles in this Declaration have been addressed. The protocol should include information regarding funding, sponsors, institutional affiliations, potential conflicts of interest, incentives for subjects, and information regarding provisions for treating and/or compensating subjects who are harmed as a consequence of participation in the research study. In clinical trials, the protocol must also describe appropriate arrangements for post-trial provisions	The 2024 version expands “incentives for participants” and adds “anticipated benefits and potential risks and burdens,” providing more transparency. It also removes “institutional affiliations,” focusing on essential details. It also sets higher requirements for information covered by program content and emphasizes the importance of privacy
23	The protocol must be submitted for consideration, comment, guidance, and approval to the concerned research ethics committee before the research begins. This committee must be transparent in its functioning and must have the independence and authority to resist undue influence from the researcher, the sponsor, or others. The committee must have sufficient resources to fulfill its duties, and its members and staff must collectively have adequate education, training, qualifications, and diversity to effectively evaluate each type of research it reviews . The committee must have sufficient familiarity with local circumstances and context, and include at least one member of the general public . It must take into consideration the ethical, legal, and regulatory norms and standards of the country or countries in which the research is to be performed as well as applicable international norms and standards, but these must not be allowed to reduce or eliminate any of the protections for research participants set forth in this Declaration. When collaborative research is performed internationally, the research protocol must be approved by research ethics committees in both the sponsoring and host countries . The committee must have the right to monitor, recommend changes to, withdraw approval for, and suspend ongoing research . Where monitoring is required, the researcher must provide information to the committee and/or competent data and safety monitoring entity, especially about any serious adverse events. No amendment to the protocol may be made without consideration and approval by the committee. After the end of the research, the researchers must submit a final report to the committee containing a summary of the findings and conclusions	The research protocol must be submitted for consideration, comment, guidance and approval to the concerned research ethics committee before the study begins. This committee must be transparent in its functioning; must be independent of the researcher, the sponsor, and any other undue influence; and must be duly qualified. It must take into consideration the laws and regulations of the country or countries in which the research is to be performed as well as applicable international norms and standards but these must not be allowed to reduce or eliminate any of the protections for research subjects set forth in this Declaration. The committee must have the right to monitor ongoing studies. The researcher must provide monitoring information to the committee, especially information about any serious adverse events. No amendment to the protocol may be made without consideration and approval by the committee. After the end of the study, the researchers must submit a final report to the committee containing a summary of the study’s findings and conclusions	The 2024 version adds requirements for the ethics committee’s “resources necessary to fulfill its duties,” specifies “local circumstances and context,” and requires a “member of the public.” It increases the degree of regulation of the organization of ethics committees. It enhances transparency and oversight, especially in international cooperation. The functions of ethics committees with more authority are emphasized: the power to request modifications to protocols for research, to withdraw information, and to suspend ongoing research, among others

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Table 1. Continued

	2024 Version text	2013 Version text	Difference analysis and significance of changes
24	Every precaution must be taken to protect the privacy of research participants and the confidentiality of their personal information	Every precaution must be taken to protect the privacy of research subjects and the confidentiality of their personal information	No change
25	Free and informed consent is an essential component of respect for individual autonomy. Participation by individuals capable of giving informed consent in medical research must be voluntary. Although it may be appropriate to consult family members or community representatives, individuals capable of giving informed consent may not be enrolled in research unless they freely agree	Participation by individuals capable of giving informed consent as subjects in medical research must be voluntary. Although it may be appropriate to consult family members or community leaders, no individual capable of giving informed consent may be enrolled in a research study unless he or she freely agrees	The 2024 version emphasizes “respect for individual autonomy,” which highlights ethical grounding, emphasizing that autonomous informed consent is an integral part of respect for the autonomy of the individual. And uses “family members or community representatives” for inclusive language
26	In medical research involving human participants capable of giving informed consent, each potential participant must be adequately informed in plain language of the aims, methods, anticipated benefits and potential risks and burdens, qualifications of the researcher , sources of funding, any potential conflicts of interest, provisions to protect privacy and confidentiality , incentives for participants, provisions for treating and/or compensating participants who are harmed as a consequence of participation, and any other relevant aspects of the research. The potential participant must be informed of the right to refuse to participate in the research or to withdraw consent to participate at any time without reprisal. Special attention should be given to the specific information and communication needs of individual potential participants as well as to the methods used to deliver the information. After ensuring that the potential participant has understood the information, the physician or another qualified individual must then seek the potential participant’s freely given informed consent, formally documented on paper or electronically . If the consent cannot be expressed on paper or electronically, the non-written consent must be formally witnessed and documented. All medical research participants should be given the option of being informed about the general outcome and results of the research	In medical research involving human subjects capable of giving informed consent, each potential subject must be adequately informed of the aims, methods, sources of funding, any possible conflicts of interest, institutional affiliations of the researcher, the anticipated benefits and potential risks of the study and the discomfort it may entail, post-study provisions, and any other relevant aspects of the study. The potential subject must be informed of the right to refuse to participate in the study or to withdraw consent to participate at any time without reprisal. Special attention should be given to the specific information needs of individual potential subjects as well as to the methods used to deliver the information. After ensuring that the potential subject has understood the information, the physician or another appropriately qualified individual must then seek the potential subject’s freely given informed consent, preferably in writing. If the consent cannot be expressed in writing, the non-written consent must be formally documented and witnessed. All medical research subjects should be given the option of being informed about the general outcome and results of the study	It added provisions for informed language to be plain and easy to understand; Richer content has been added to participant presentations, including information on possible burdens, privacy protection, potential harms of participation, and compensation for any harm incurred; Informed content has been added, specifying that participants have the right to refuse participation and to withdraw their information at any time; Attention to detail has been added regarding the special needs of certain participants in the communication of information; It also emphasizes the importance of the methods and ways of communication. The inclusiveness of e-drafts has also been enhanced
27	When seeking informed consent for participation in research, the physician or other researcher must be particularly cautious if the potential participant is in a dependent relationship with them or may consent under duress. In such situations, the informed consent must be sought by an appropriately qualified individual who is independent of this relationship	When seeking informed consent for participation in a research study, the physician must be particularly cautious if the potential subject is in a dependent relationship with the physician or may consent under duress. In such situations, the informed consent must be sought by an appropriately qualified individual who is completely independent of this relationship	The 2024 version includes “other researcher” alongside the physician and adopts “independent of this relationship” to maintain consent integrity

(Continued)

Table 1. Continued

	2024 Version text	2013 Version text	Difference analysis and significance of changes
28	In medical research involving human participants incapable of giving free and informed consent, the physician or other qualified individual must seek informed consent from the legally authorized representative, considering preferences and values expressed by the potential participant . Those persons incapable of giving free and informed consent are in situations of particular vulnerability and are entitled to the corresponding safeguards . In addition to receiving the protections for the particularly vulnerable, those incapable of giving consent must only be included if the research is likely to either personally benefit them or if it entails only minimal risk and minimal burden	For a potential research subject who is incapable of giving informed consent, the physician must seek informed consent from the legally authorized representative. These individuals must not be included in a research study that has no likelihood of benefit for them unless it is intended to promote the health of the group represented by the potential subject, the research cannot instead be performed with persons capable of providing informed consent, and the research entails only minimal risk and minimal burden	In the original "legal representative" signing section, "the personal preferences and values of potential participants should also be considered" was added; it was re-emphasized that those who cannot give informed consent on their own are a vulnerable group and should be protected; the scope of circumstances under which people who cannot give informed consent can be included as participants in a study were revised, narrowed down, and made more stringent
29	When a potential research participant who is incapable of giving free and informed consent is able to give assent to decisions about participation in research, the physician or other qualified individual must seek that assent in addition to the consent of the legally authorized representative, considering any preferences and values expressed by the potential participant . The potential participant's dissent should be respected	When a potential research subject who is deemed incapable of giving informed consent is able to give assent to decisions about participation in research, the physician must seek that assent in addition to the consent of the legally authorized representative. The potential subject's dissent should be respected	The 2024 version includes "other qualified individuals" and adds "preferences and values," ensuring ethical care
30	Research involving participants who are physically or mentally incapable of giving consent (e.g. unconscious patients) may be done only if the physical or mental condition that prevents giving informed consent is a necessary characteristic of the research group. In such circumstances, the physician or other qualified individual must seek informed consent from the legally authorized representative. If no such representative is available and if the research cannot be delayed, the research may proceed without informed consent, provided that the specific reasons for involving participants with a condition that renders them unable to give informed consent have been stated in the research protocol and the research has been approved by a research ethics committee. Free and informed consent to remain in the research must be obtained as soon as possible from a legally authorized representative or, if they regain capacity to give consent , from the participant	Research involving subjects who are physically or mentally incapable of giving consent, e.g. unconscious patients, may be done only if the physical or mental condition that prevents giving informed consent is a necessary characteristic of the research group. In such circumstances, the physician must seek informed consent from the legally authorized representative. If no such representative is available and if the research cannot be delayed, the study may proceed without informed consent provided that the specific reasons for involving subjects with a condition that renders them unable to give informed consent have been stated in the research protocol and the study has been approved by a research ethics committee. Consent to remain in the research must be obtained as soon as possible from the subject or a legally authorized representative	The 2024 version expands the scope to include "other qualified individuals" other than physicians and clarifies the process for re-consent, emphasizing that consent must be autonomous, informed, and capable of informed consent
31	The physician or other researcher must fully inform potential participants which aspects of their care are related to the research. The refusal of a patient to participate in research or the patient's decision to withdraw from research must never adversely affect the patient-physician relationship or provision of the standard of care	The physician must fully inform the patient which aspects of their care are related to the research. The refusal of a patient to participate in a study or the patient's decision to withdraw from the study must never adversely affect the patient-physician relationship	The 2024 version adds "other researchers" to broaden the scope of research responsibilities and emphasize their inclusiveness. It also expands the scope of protection for participants who withdraw from a trial, further safeguarding the interests of participants

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Table 1. Continued

	2024 Version text	2013 Version text	Difference analysis and significance of changes
32	Physicians or other qualified individuals must obtain free and informed consent from research participants for the collection, processing , storage, and foreseeable secondary use of biological material and identifiable or re-identifiable data. Any collection and storage of data or biological material from research participants for multiple and indefinite uses should be consistent with requirements set forth in the WMA Declaration of Taipei, including the rights of individuals and the principles of governance. A research ethics committee must approve the establishment and monitor ongoing use of such databases and biobanks. Where consent is impossible or impracticable to obtain, secondary research on stored data or biological material may be done only after consideration and approval of a research ethics committee	For medical research using identifiable human material or data, such as research on material or data contained in biobanks or similar repositories, physicians must seek informed consent for its collection, storage, and/or reuse. There may be exceptional situations where consent would be impossible or impracticable to obtain for such research. In such situations, the research may be done only after consideration and approval of a research ethics committee	The term "free and informed" was introduced; a link to the processing of human material or data was added; the original "In exceptional cases, informed consent may be dispensed with after obtaining ethical approval" was deleted; "Informed consent may be dispensed with after obtaining ethical approval" was deleted; "Informed consent may be dispensed with after obtaining ethical approval" was deleted; in its place, the WMA Taipei Declaration is to be complied with, and the approval of the Ethics Committee is required, and the Ethics Committee is required to monitor the process of use
33	The benefits, risks, burdens, and effectiveness of a new intervention must be tested against those of the best proven intervention(s), except in the following circumstances: If no proven intervention exists, the use of placebo, or no intervention, is acceptable; or If for compelling and scientifically sound methodological reasons the use of any intervention other than the best proven one(s), the use of placebo, or no intervention is necessary to determine the efficacy or safety of an intervention; and the participants who receive any intervention other than the best proven one(s), placebo, or no intervention will not be subject to additional risks of serious or irreversible harm as a result of not receiving the best proven intervention. Extreme care must be taken to avoid abuse of this option	The benefits, risks, burdens, and effectiveness of a new intervention must be tested against those of the best proven intervention(s), except in the following circumstances: Where no proven intervention exists, the use of placebo, or no intervention, is acceptable; or Where for compelling and scientifically sound methodological reasons the use of any intervention less effective than the best proven one, the use of placebo, or no intervention is necessary to determine the efficacy or safety of an intervention and the patients who receive any intervention less effective than the best proven one, placebo, or no intervention will not be subject to additional risks of serious or irreversible harm as a result of not receiving the best proven intervention. Extreme care must be taken to avoid abuse of this option	The 2024 version clarifies criteria, replacing "less effective than the best proven one" with "any intervention other than the best proven one(s)," resulting in a more accurate presentation
34	In advance of a clinical trial, post-trial provisions must be arranged by sponsors and researchers to be provided by themselves , healthcare systems, or governments for all participants who still need an intervention identified as beneficial and reasonably safe in the trial. Exceptions to this requirement must be approved by a research ethics committee. Specific information about post-trial provisions must be disclosed to participants as part of informed consent	In advance of a clinical trial, sponsors, researchers, and host country governments should make provisions for post-trial access for all participants who still need an intervention identified as beneficial in the trial. This information must also be disclosed to participants during the informed consent process	Responsible parties were expanded to include the sponsor, the investigator, the medical team, and the government; the importance of informed consent was also emphasized by the requirement that only valid and safe participants must be provided with a continuous supply of the trial drug after the trial ends
35	Medical research involving human participants must be registered in a publicly accessible database before recruitment of the first participant	Every research study involving human subjects must be registered in a publicly accessible database before recruitment of the first subject	The 2024 version updates the terminology from "human subjects" to "human participants" to reflect more respectful language

(Continued)

Table 1. Continued

	2024 Version text	2013 Version text	Difference analysis and significance of changes
36	Researchers, authors, sponsors, editors, and publishers all have ethical obligations with regard to the publication and dissemination of the results of research. Researchers have a duty to make publicly available the results of their research on human participants and are accountable for the timeliness, completeness, and accuracy of their reports. All parties should adhere to accepted guidelines for ethical reporting. Negative and inconclusive as well as positive results must be published or otherwise made publicly available. Sources of funding, institutional affiliations, and conflicts of interest must be declared in the publication. Reports of research not in accordance with the principles of this Declaration should not be accepted for publication When an unproven intervention is utilized in an attempt to restore health or alleviate suffering for an individual patient because approved options are inadequate or ineffective and enrollment in a clinical trial is not possible, it should subsequently be made the object of research designed to evaluate safety and efficacy. Physicians participating in such interventions must first seek expert advice, weigh possible risks, burdens, and benefits, and obtain informed consent. They must also record and share data when appropriate and avoid compromising clinical trials. These interventions must never be undertaken to circumvent the protections for research participants set forth in this Declaration	Researchers, authors, sponsors, editors, and publishers all have ethical obligations with regard to the publication and dissemination of the results of research. Researchers have a duty to make publicly available the results of their research on human subjects and are accountable for the completeness and accuracy of their reports. All parties should adhere to accepted guidelines for ethical reporting. Negative and inconclusive as well as positive results must be published or otherwise made publicly available. Sources of funding, institutional affiliations, and conflicts of interest must be declared in the publication. Reports of research not in accordance with the principles of this Declaration should not be accepted for publication In the treatment of an individual patient, where proven interventions do not exist or other known interventions have been ineffective, the physician, after seeking expert advice, with informed consent from the patient or a legally authorized representative, may use an unproven intervention if in the physician's judgment it offers hope of saving life, re-establishing health, or alleviating suffering. This intervention should subsequently be made the object of research, designed to evaluate its safety and efficacy. In all cases, new information must be recorded and, where appropriate, made publicly available	The 2024 version introduces the term "timeliness," enhancing accountability for prompt reporting
37			The condition of "not proven effective in clinical practice" was limited to "inability to enroll in relevant clinical trials of the intervention" in addition to the absence of other proven treatments; the purpose of the trial was retained as "to restore health and alleviate suffering"; the phrase "only after expert approval" was separated into a second sentence, which, in addition to being emphasized, also added the requirement that the risks, burdens, and benefits should be properly balanced. "Restoration of health and relief of suffering" was retained; the need for a good balance of risks, burdens, and benefits was emphasized; and the description of the application of the intervention required that data be documented and shared in a way that did not interfere with ongoing clinical trials of the intervention

for ethics committee members underscore the importance of culturally and contextually informed oversight, ensuring that all research aligns with local social and ethical standards. For instance, in multinational trials for diseases like tuberculosis, dual ethical reviews help balance local cultural norms with international standards, prioritizing participants' welfare across diverse regulatory landscapes. Additionally, the WMA promotes the establishment and standardization of ethics committees globally to enhance the precision and efficiency of ethical review. The responsibilities of ethics committees have also been extended from initial review to ongoing supervision and data monitoring, further ensuring data security and standardization in the research process [9].

Engaging participants and communities

The updated Declaration advocates for meaningful engagement with participants and their communities throughout the research lifecycle. Researchers are encouraged to interact meaningfully with participants and communities before, during, and after conducting medical research. This engagement allows participants and communities to share their priorities and values, contribute to study design and conduct, and participate in understanding and disseminating results. This principle promotes transparency, trust, and shared decision-making. For example, recent oncology studies have demonstrated the value of patient advocacy group involvement, leading to studies that address practical concerns relevant to cancer patients, such as quality of life and side effect management. By incorporating community perspectives, researchers can ensure that studies align closely with the needs and values of those they serve, enhancing ethical integrity and study relevance. The Declaration also mandates post-trial safeguards for participants needing continued intervention, ensuring that beneficial treatments discovered in trials are not discontinued at the trial's conclusion without ethics committee approval. This measure promotes post-trial equity and supports participant well-being [10, 11].

Protection and inclusion of vulnerable populations

Since its inception, the WMA has emphasized vulnerability in the Declaration. The 2024 revision expands this concept to encompass not only individual vulnerability but also community and population vulnerability. This expansion aims to provide disadvantaged populations—such as the elderly, women, children, refugees, the chronically ill, and those in economically underdeveloped areas—with equal opportunities to participate in and benefit from medical research. It highlights a nuanced understanding of social determinants of health. For example, research among refugee populations illustrates the importance of balancing representation in clinical studies with protections against potential exploitation. This expansion acknowledges the interconnectedness of individual and communal health and seeks to prevent disparities in research from reinforcing existing inequalities. This shift aligns with the Declaration's patient-centered approach and reaffirms the centrality of human welfare in medical research [12].

Data privacy and biobank consent

Since 2016, the Declaration has been supplemented by the Taipei Declaration on Ethical Considerations for Health Databases and Biobanks. However, with the widespread application of AI, machine learning, and big data in medical research, the Declaration's original data management guidelines have become insufficient to address contemporary experimental scenarios.

Particularly, the current provisions contain loopholes regarding data confidentiality and participant privacy protection. The 2024 revision addresses these gaps by strengthening requirements for explicit consent on secondary uses of data and stored biological materials. For example, AI-driven analyses of genomic data in biobanks highlight the need for clear participant consent regarding future uses, as potential re-identification of anonymized data could lead to unintended privacy breaches. Ethics committees are now required to oversee these practices, adding a protective layer to ensure that data use aligns with participants' rights and expectations [13].

Public health emergency ethics

The COVID-19 pandemic underscored the ethical challenges of conducting research during public health crises. Paragraph 8 of the 2024 Declaration reinforces the necessity of adhering to ethical standards, even in emergencies, challenging the notion of "pandemic exceptionalism." During COVID-19, accelerated vaccine trials raised concerns over informed consent, as rapid enrollment processes occasionally left participants underinformed about potential risks. This revision asserts that urgency does not negate the ethical obligation to protect participants, promoting scientific rigor and transparency, even in crises. This update defines and constrains ethical boundaries for conducting medical research in emergencies, balancing research efficiency with participant protection, and providing a strong theoretical foundation for future ethical reviews in similar situations.

Environmental sustainability

In alignment with environmental responsibility, the 2024 Declaration now mandates that research practices minimize ecological impacts. For example, the energy demands of data processing centers integral to large-scale trials underscore the relevance of this addition. Research organizations are encouraged to adopt energy-efficient data management solutions, such as low-energy servers, to align medical research with broader environmental goals. The Declaration also discourages redundant experimental designs and studies that may not fully meet ethical review standards, thus maintaining both the integrity and sustainability of research practices. This expansion broadens the scope of ethical responsibility, linking human health with environmental stewardship and reflecting the framers' and revisers' intent to elevate ethical standards while maintaining a patient-centered focus that considers the ecological and social impacts of medical research.

Discussion and conclusion

The 2024 revision of the Declaration of Helsinki represents a significant ethical milestone, blending traditional principles of participant protection with adaptations to address contemporary challenges in clinical research. This update reinforces a commitment to patient-centered research ethics by promoting respect, inclusivity, and transparency, recognizing participants as active partners in the research process. Expanded guidelines on community engagement, environmental responsibility, and dual-layered ethical reviews highlight the Declaration's responsiveness to modern research complexities, establishing a robust ethical standard that aligns with societal expectations and scientific mandates for participant protection.

However, despite these advancements, areas for further ethical guidance remain. For example, the rapid adoption of AI in

clinical research introduces complex ethical dilemmas that the current Declaration only partially addresses with respect to data safety. As AI increasingly influences trial design, data analysis, and decision-making, issues such as data bias, algorithmic transparency, and patient autonomy will require clearer ethical guidelines to prevent unintentional harm. Incorporating AI-specific guidelines would better prepare the Declaration for these challenges, ensuring it remains relevant and effective in safeguarding participants' rights as research methodologies evolve.

Moreover, additional elements should be considered for the Declaration's framework on medical research ethics during emergency situations. Specifically, guidelines could further explore how the Declaration's principles can serve as ethical safeguards rather than impediments to research, particularly in crisis contexts. Questions around conducting ethical reviews during emergencies and adjusting review rigor based on situational realities to ensure experimental efficiency remain unaddressed and need to be tested in practice.

The scope of the Declaration's application is another area for future consideration. Currently, the Declaration is limited to human research, excluding fields like forensic science and human specimen research, which lack comprehensive ethical standards for oversight. Although the 2008 amendment addressed informed consent for human tissues, the practicality and regulatory effectiveness of this section have faced criticism [14]. Expanding the Declaration to cover human specimen research or developing a new set of principles for these fields would enhance its regulatory relevance and ethical reach [15].

Looking ahead, the Declaration must continue adapting to new and unforeseen ethical challenges posed by technological advances. Future revisions should prioritize ethical frameworks for AI and data-driven methodologies, alongside refined protocols for protecting vulnerable populations, especially within globalized and diverse research settings. By incorporating these considerations, the Declaration can remain a foundational document championing respect, responsibility, and scientific integrity in the medical research community.

This revised Declaration, strengthened by these perspectives, is positioned to guide ethical research into a new era. As the biomedical field evolves, so too must the ethical principles that guide it, ensuring that all research participants are protected and respected across every context and technological landscape.

Author contributions

B.W. and G.Z. conceived the idea and wrote the first draft, tables, and figures. C.Z., C.C., and H.Y. made substantial revisions to the draft. All authors contributed to the revisions and approved the final version.

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